

How much neighbourhood detail do published statistics contain?

An exchange rate between aggregate social data and satellite maps

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Abstract

Most countries publish social statistics as administrative aggregates and almost none publish them at neighbourhood level. We measure how much neighbourhood detail those aggregates contain. Training a weakly supervised model on nothing but five-level deprivation grades of Mexican municipalities and free satellite imagery, and validating against fully held-out census-tract grades, we find that municipal aggregates recover 80% of the tract-level ordering attainable with full spatial supervision (within-municipality $\rho = 0.213$ of a 0.266 ceiling). One hundred aggregates already recover most of it; coarsening the label to state level halves it, and to one national figure erases it. The resulting maps close 26–41% of the gap between aggregate-guided and census-guided targeting of social programmes, replicate against an independent institution’s index ($\rho = 0.37$), outperform the leading downloadable wealth map, and transfer without retraining to Bogotá ($\rho = 0.38$ against block-level strata). Popular attention-based multiple-instance models fail at this task for a measurable reason we diagnose and fix.

Introduction

Census-quality social data is expensive, and most of the world sees it roughly once a decade. In between, and below the level the tables are published at, policy runs on aggregates: a municipal index here, a district average there. A rich line of work has shown that satellite imagery plus machine learning can predict such indicators across space [1–4], and global products derived from that idea are already used for programme targeting [3, 5]. But prediction across places and *disaggregation within them* are different tasks, and the second one carries the policy weight: a mayor knows their municipality is poor; what they lack is the map of which neighbourhoods are poorest.

This paper asks the disaggregation question quantitatively: *how much neighbourhood detail does a published aggregate contain?* We treat the answer as an exchange rate — so many aggregates of such a size buy so much map — and we measure it where it can be measured: Mexico publishes its Social Deprivation Grade (GRS) both as a municipal aggregate and at census-tract

level (AGEB) for 61,430 urban tracts [6]. We train only on the former and validate only on the latter, held out in full, on cities the model never saw.

Three design choices make the measurement clean. First, supervision is genuinely weak: a model sees a bag of image tokens for a municipality and one label — the shares of population at each deprivation grade — and never any spatial annotation. Second, every comparison is scored *within* municipalities, because ordering tracts across cities is mostly solved by knowing which city is poor, and pooled metrics flatter every method (pooled $\rho = 0.44$ against 0.21 within). Third, an oracle trained on the held-out tract labels bounds what the frozen representation supports at all; every result is reported as a fraction of that ceiling, so “how good is weak supervision” is separated from “how good are the features”.

Results

Attention does not localize, and the reason is measurable

We first trained the standard tool for weakly supervised localization: gated-attention multiple-instance learning (ABMIL [7], with the CLAM instance constraint [8]). It predicts municipal labels well (quadratic-weighted $\kappa = 0.62$) and its attention map is at chance (AUROC 0.463 for recovering high-deprivation tracts; Fig. 2a). The diagnosis is quantitative: flattening the attention to near-uniform via an entropy penalty costs no bag accuracy (AUROC $0.899 \rightarrow 0.895$), so the bag task is solvable from the plain feature mean and no gradient ever asks the attention to point anywhere. Four ablations (ordinal and cumulative labels, clustering constraints, entropy sweeps) confirm the pattern.

Inverting the composition fixes it. Predicting every token first and averaging the *predictions* — learning from label proportions (LLP [9]) — makes the map the mechanism rather than a by-product: the same features, bags and labels reach AUROC 0.761. Because Mexican municipal grades are literally population-weighted aggregates of tract grades, the setting is LLP with known instance weights, and using census population as those weights improves the map further (within-municipality ρ $0.213 \rightarrow 0.234$, bag error $0.090 \rightarrow 0.087$). A hyperparameter sweep — chosen on training folds only, under the one signal weak supervision may legitimately monitor — accounts for most of the remaining gap to the ceiling (Fig. 2a).

The sweep exposed a structural blindness worth naming. Across fourteen configurations, the rank correlation between bag-level error (the honest model-selection criterion) and map quality is -0.14 : selecting weakly supervised models by the only metric they may look at is nearly blind to the quantity that matters (Fig. 2b). This applies to any pipeline whose product is the map — computational pathology included — and it is why we release our benchmark with held-out instance labels.

The exchange rate: three axes

Figure 1 shows the headline measurement on an expanded pool of 1,144 municipal bags covering essentially all of urban Mexico. **Quantity** (Fig. 1a): fifty aggregates buy $\rho = 0.15$, one hundred buy 0.18, and the full national supply buys 0.213 — 80% of the 0.266 oracle ceiling. The curve’s knee is early: most of the attainable map exists with a few hundred published numbers.

Granularity (Fig. 1b): re-labelling the same bags with state-level aggregates halves the map ($\rho = 0.119$); a single national aggregate erases it (0.004). What a statistics office chooses to publish is, quite literally, the map its users can recover. **Resolution** (Fig. 1c): destroying the optical resolution to the radar’s 20 m by block-averaging barely hurts ($\rho = 0.183 \rightarrow 0.172$), while radar itself collapses to chance (0.006) despite being the best single modality for *predicting* municipal grades. Content, not ground sampling distance, is what fails: the dissociation between predicting and localizing recurs across sensors, not just across models.

External validity

Four independent checks argue the map is measurement rather than artifact (Fig. 3). It outperforms the leading downloadable product — Meta’s Relative Wealth Index [3] — under our exact protocol ($\rho = 0.107 \pm 0.082$ vs 0.213; Fig. 3a), and a land-cover-plus-NDVI head with no learned representation reaches 0.149, so the foundation-model features carry real signal beyond built-up density. It replicates against an institution it never trained toward: CONAPO’s continuous urban marginalization index, built from a different indicator set, correlates at $\rho = 0.366 \pm 0.073$ within municipalities — higher than against the five-level training construct, as a tie-free target should be (Fig. 3b). In simulated geographic targeting, allocating a fixed budget by the map instead of by the municipal aggregate closes 26–41% of the gap to census-guided allocation, growing with budget (Fig. 3c). And a border-discontinuity control finds no administrative short-cut: crossing a municipal border where deprivation does not change moves the map by $+0.013$ $[-0.006, +0.247]$ once neighbourhood context is computed city-wide — an earlier $+0.143$ jump traced to bag-limited adjacency, our seam rather than the model’s — while tract borders inside municipalities are crossed smoothly (-0.002).

The construct travels; a different construct does not

Applied unchanged to Bogotá, the Mexican model orders Colombia’s block-level socioeconomic strata at $\rho = 0.378$ (AUROC 0.732 for strata 1–2; Fig. 4). Applied to Rio de Janeiro, it detects IBGE’s mapped informal settlements barely above chance (AUROC 0.581 among built-up tokens). The contrast is informative: Colombian strata measure the same construct as the GRS — a graded deprivation ordering over the whole urban fabric — while the AGSN mask delimits informality, a partly orthogonal notion. Weak supervision on aggregates taught the model a deprivation gradient, and that gradient, not a country signature, is what transfers.

Limits

The ceiling itself is the honest limit: even with full tract labels and neighbourhood context, 160 m tokens support $\rho = 0.266$ within municipalities. Agreement rises steeply with tract size (from 0.18 for tracts of 1–3 tokens to 0.45 for 20–49; Supplementary Table) — the modifiable areal unit problem stated in units an office can act on. Maps at this fidelity order neighbourhoods usefully for prioritization and do not read individual blocks.

Discussion

The exchange rate reframes a familiar technology. Rather than another demonstration that imagery predicts poverty, it answers the question a statistics office would actually ask: *what resolution of publication buys what resolution of inference?* The three axes give the answer an operational shape — a few hundred aggregates suffice; publish municipal rather than state tables; optical resolution is not the binding constraint — and the targeting experiment prices it in people reached. The negative results carry equal weight: attention-based MIL fails here for a structural reason, aggregate-level validation is nearly blind to map quality, and informality is not a free by-product of a deprivation gradient. All three are warnings any weakly supervised mapping effort will meet.

Mexico’s dual-level publication made the calibration possible; the measured fractions are what travel to the majority of countries that publish only aggregates. The Bogotá result is the first step of that argument, and extending the dual calibration to Colombia’s own strata is the natural next measurement.

Methods

Data. Sentinel-2 L2A and Sentinel-1 RTC annual median composites for 2020 over 424 Mexican cities and municipalities (10 m grid, windowed reads from Planetary Computer), CONEVAL GRS at municipal and AGEB level (2020 census), CONAPO IMU 2020, ESA WorldCover 2020, Meta RWI, IBGE AGSN 2019, Bogotá block stratification. **Bags and instances.** A bag is a municipality; instances are 16-pixel tokens (160 m) encoded by a frozen DOFA foundation model [10]; both sensors are fused per token. Labels are the population shares at each grade threshold. **Model.** A two-layer head over frozen features predicts four cumulative-grade probabilities per token; the bag prediction is the (population-weighted) mean of token predictions, fit with binary cross-entropy on standardized inputs; each token’s hidden representation is concatenated with the mean of its 8 grid neighbours (480 m). **Protocol.** 110 training / 14 validation / 14 test cities, stratified, no city split across sets; the test set remains unopened. All sweeps select on grouped training folds; validation cities are scored once per reported configuration; comparisons are paired by municipality over three seeds; intervals are city-clustered bootstraps. **Oracle.** The same head trained directly on tract labels bounds the representation. **Code and data.** Full pipeline, frozen vectors, labels, splits and protocol released as a benchmark.

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Figures

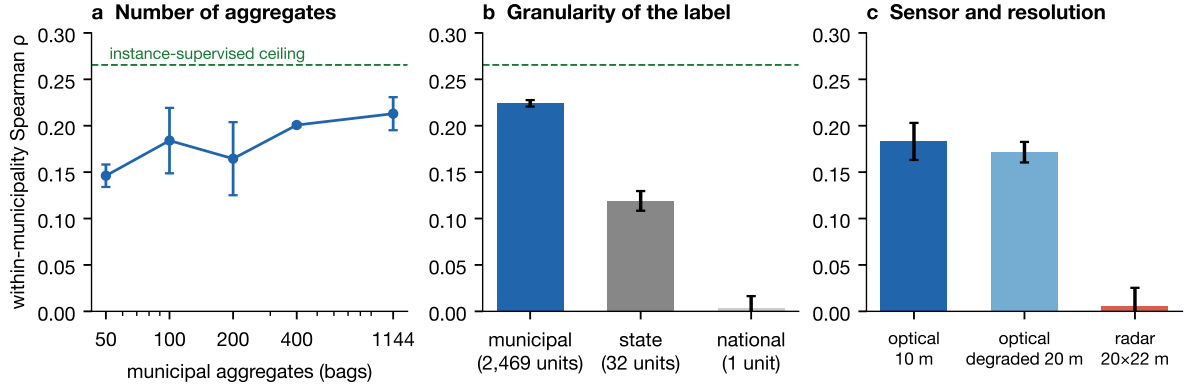


Figure 1: **The exchange rate between aggregates and maps.** **a**, Within-municipality agreement between the weakly supervised map and held-out tract grades, as a function of how many municipal aggregates supervise training (nested, grade-stratified sampling; mean $\pm$ s.d. over three seeds; dashed line: oracle trained on tract labels). **b**, The same bags re-labelled with coarser aggregates. **c**, Sensor and resolution: block-averaging optical imagery to the radar’s resolution barely hurts; radar collapses despite being the best predictor of municipal grades.

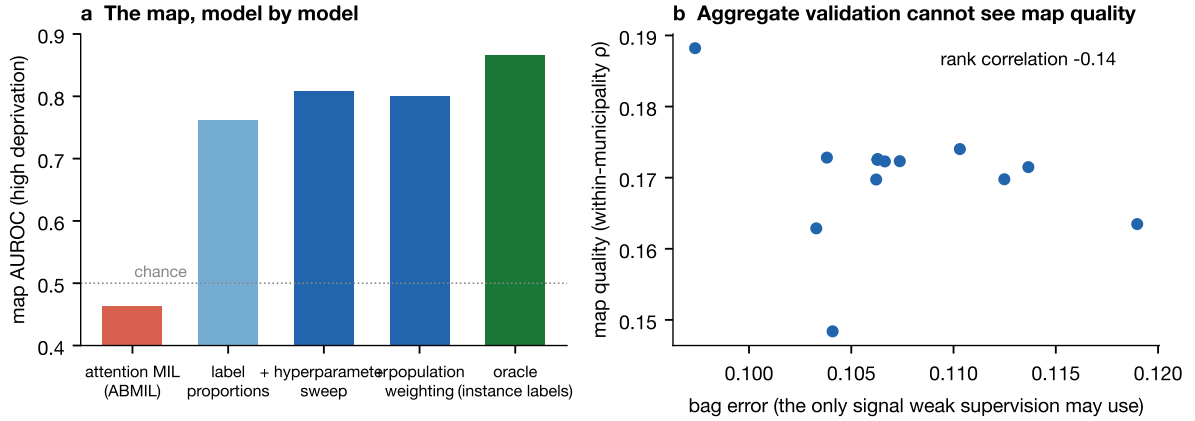


Figure 2: **Predicting is not localizing.** **a**, Map quality by model family on identical features, bags and labels. Attention-based MIL is at chance because the bag task is solvable from the feature mean; predicting instances and averaging predictions (label proportions) makes the map the mechanism. **b**, Across the hyperparameter sweep, the only criterion weak supervision may monitor (bag error) is nearly uncorrelated with map quality.

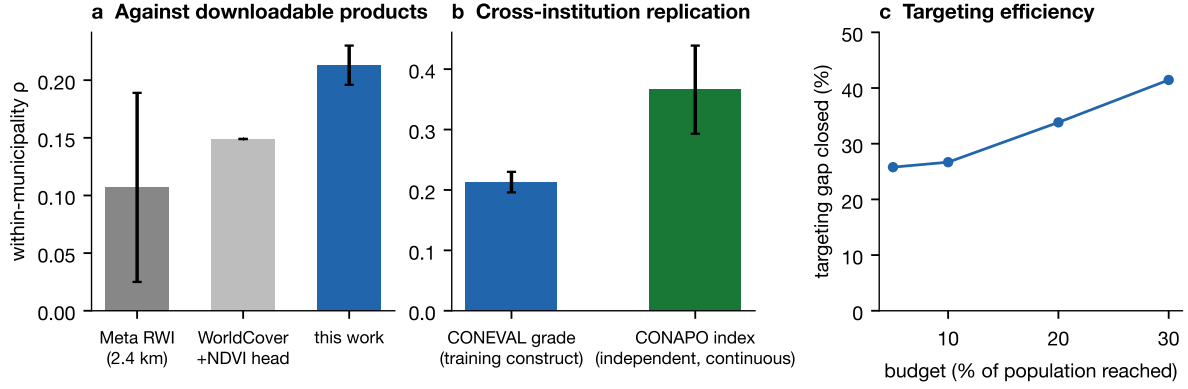


Figure 3: **External validity.** **a**, Within-municipality agreement of downloadable products and of this work, under one protocol. **b**, Replication against an independent institution’s continuous index. **c**, Share of the aggregate-to-census targeting gap closed by allocating a fixed budget with the map (people pooled across the 14 held-out cities).

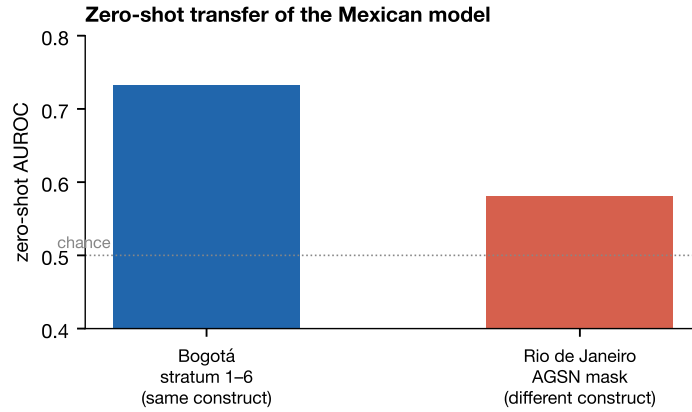


Figure 4: **Zero-shot transfer.** The Mexican model, unchanged, ordering Bogotá’s block-level strata and detecting Rio’s mapped informal settlements. The deprivation gradient travels; informality is a different construct.